William Cheung

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EXPERIENCE

Squarespace, New York, NY — **Senior Data Scientist** July 2025 – Present

Marketing Science

- Build and calibrate Bayesian Marketing Mix Models across multiple geos and channels (Hill saturation curves, spend optimization) and lead the evaluation of Google Meridian for adoption (Python, NumPyro, SciPy)
- Own multi-touch attribution pipelines, integrating a paid media partner's touchpoint data and A/B and geo-test results into incrementality-adjusted channel weights (BigQuery, dbt, Airflow)
- Productionized MMM outputs into warehouse tables and recurring weight updates that feed channel-level attribution, and prototyped a Media Measurement Hub so marketing stakeholders can run budget-allocation scenarios (BigQuery, dbt, Hex)
- Improved trust in the LTV metric used by marketing and finance through segment-level LTV computation, imputation fixes, and data-quality monitoring on subscription score tables

Agent / LLM Evals

- Lead evaluation design for an internal NL-to-SQL analytics agent, combining an LLM-as-judge rubric, deterministic SQL correctness checks, and a golden benchmark dataset (DeepEval, BigQuery)
- Built an NLG scoring service (FastAPI, 7 endpoints, batch scoring) packaged as an installable library reused across multiple agent projects (Python, Gemini)
- Implemented offline batch evals, online eval gates, and pre-call / pre-execute / post-execute guardrails in shadow mode, plus a confidence score for generated queries (LangGraph, Airflow)

Goodway Group, New York, NY — **Senior Data Scientist** April 2023 – July 2025

- Led development of media buying algorithms for pricing and volume control; benchmarked against The Trade Desk Koa and achieved a 23% reduction in CPM with no significant impact on CPA
- Built internal Marketing Mix Modeling tools and a scenario-modeling extension with model storage in MLflow, driving a 17–53% increase in ROAS across clients
- Partnered with product, engineering, and client teams to deploy Streamlit apps (A/B testing framework, brand assessment, Daido chatbot), reaching up to 45% usage among client teams
- Created internal A/B testing processes for back-end optimization and guided analytics teams in launching client-side tests

Criteo, New York, NY — **Data Scientist** May 2021 – April 2023

- Built self-service reporting and performance insights for online video campaigns (OLV, OTT), helping grow client spend ~300% year over year on upper- and full-funnel tactics
- Built A/B testing infrastructure integrating internal processes with a newly acquired platform (IponWeb), enabling internal teams to run their own tests
- Applied NLP to classify app inventory type, improving reporting and training data for the bidding engine

Fordham University, New York, NY — **Graduate Researcher (Machine Learning)** Sept 2019 – May 2021

- Built continuous authentication models for IoT wearables from heart rate, gait, and breathing data (RF, SVM, CNN, RNN) reaching 97% accuracy, deployed end to end via Fitbit, Android, and server apps; 2 IEEE publications

National Grid, Brooklyn, NY — **Financial Analyst** June 2016 – Sept 2019

- Led month-end close analysis for Narragansett Electric, establishing cost-capture methods for recovery of over $10B, and automated batch entries with VBA/SAP, cutting the close by one workday

TECHNICAL SKILLS

- **Languages:** Python (pandas, NumPy, SciPy, scikit-learn, NumPyro), SQL (BigQuery), R, Bash
- **Machine Learning & Statistics:** Machine learning (regression, classification, NLP), causal

inference, Bayesian modeling, statistical modeling, time series forecasting
- **Marketing Science & Experimentation:** Marketing Mix Modeling (Google Meridian), multi-touch attribution, incrementality and geo-testing, A/B testing, LTV modeling
- **LLM / Agents:** LangGraph, DeepEval, LLM-as-judge evaluation, Gemini, FastAPI
- **Data & MLOps:** dbt, Airflow, Google Cloud Platform, AWS, MLflow, Git
- **Visualization & Apps:** Hex, Streamlit, Plotly

EDUCATION

Fordham University — M.S. in Data Science (GPA: 4.0) Dec 2020
Stony Brook University — B.S. in Applied Mathematics and Economics, double major (GPA: 3.6) May 2016

PUBLICATIONS

- Cheung, W. and Vhaduri, S. "Continuous Authentication of Wearable Device Users from Heart Rate, Gait, and Breathing Data." IEEE BioRob. arXiv:2008.10779
- Cheung, W. and Vhaduri, S. "Context-Dependent Implicit Authentication for Wearable Device User." IEEE PIMRC. arXiv:2008.12145